Qn. Set Code-7 (SOT)

Semester: 2nd
Programme: M.Tech/Ph.D
Branch(Specialization):
Comp.Sc.CE(CSE,DA,IS,SE)

SPRING END SEMESTER EXAMINATION-2024

2nd Semester M.Tech / Ph.D Coursework

COMPUTATIONAL INTELLIGENCE

CS-60102

(For 2023 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions: [1 × 10]

(a) Classify the following statement as true (T) or false (F). Justify.

A fuzzy set is a crisp set but the reverse is not true.

(b) Define the advantages of Genetic algorithms (GAs) over conventional algorithms.

(c) Match the following: GA, Fuzzy logic, and ANN can be applied to carry out following activities respectively:

i) Classification, Optimization, Inferencing
ii) Optimization, Inferencing, Classification
iii) Inferencing, Classification, Optimization
iv) Classification, Inferencing, Optimization

(d) Suppose a fuzzy set $\bar{A}$ is described as $bell(x; a, b, c)$. Illustrate analytically that the classical fuzzy complement $\bar{A}^c$ is $bell(x; a, -b, c)$.

(e) Choose the correct option:

Which of the following statements about Particle Swarm Optimization (PSO) is true?

i) PSO is only suitable for convex optimization problems
ii) PSO guarantees finding the global optimum for any optimization problem
iii) PSO can get trapped in local optima
iv) PSO is insensitive to parameter tuning

(f) Consider a fuzzy set A defined on the interval $x \in [0, 9]$ of integers by the membership function, $\mu_{A(x)} = 3x/(5x + 2)$. Show the value of $A_{0.5}$ from the following:

i) $\{0, 2, 4, 6, 8\}$
ii) $\{1, 3, 5, 7, 9\}$
iii) $\{2, 3, 4, 5, 6, 7, 8, 9\}$
iv) $\{2, 3, 5, 7\}$

(g) Classify the following statement as true (T) or false (F). Justify.

Genetic Algorithms (GAs) are capable of handling non-differentiable and discontinuous objective functions.

(h) Relate one difference between Mamdani FIS and other two FISs (i.e. Sugeno FIS and Tsukamoto FIS):

i) Sugeno FIS and Tsukamoto FIS are less interpretable but more accurate.
ii) Mamdani FIS is easy to interpret but less accurate.
iii) Mamdani FIS needs defuzzification module whereas other two do not.
iv) Sugeno FIS and Tsukamoto FIS do not require any fuzzification module whereas Mamdani FIS needs.

(i) Define the role of the bias term in an Artificial Neural Network.

(j) Choose the correct option:

What is the primary advantage of ACO over traditional optimization algorithms?

i) ACO guarantees finding the global optimum in a finite number of iterations.
ii) ACO can handle complex, combinatorial optimization problems efficiently.

iii) ACO does not require any parameter tuning.

iv) ACO is deterministic and converges quickly.

SECTION-B

2. (a) Consider a fuzzy set $\bar{A}$, whose membership function is given by: [5]

$$\mu_{\bar{A}}(x) = \begin{cases} 1 - 2|x-1|, \text{for } |x-1| \leq 0.5. \\ 0, \quad \text{otherwise.} \end{cases}$$

(i) Construct (Sketch) the membership function.

(ii) Identify the bandwidth of $\bar{A}$.

(iii) Identify the $\alpha(= 0.25)$ cut of $\bar{A}$.

(b) Explain Linear Separability using an example. Illustrate *XOR* classifier with perceptron. [5]

3. (a) (i) Two fuzzy sets $\bar{P}$ and $\bar{Q}$ are defined (on Universal set X where $x \in X$) as follows: [5]

	x_1	x_2	x_3	x_4
$\bar{P}$	0.1	0.2	0.7	0.4
$\bar{Q}$	0.8	0.6	1.0	0.3

Solve $(\bar{P} \times \bar{Q})_{0.3}$.

(ii) Identify the fuzzy rule "IF x is $\bar{A}$ THEN y is $\bar{B}$", where fuzzy sets

$\bar{A} = \{(a, 0.1), (b, 0.9), (c, 0.8), (d, 0.3)\}$ and

$\bar{B} = \{(1, 0.3), (2, 0.7), (3, 1.0), (4, 0.5)\}$

defined on universe of discourses $X = \{a, b, c, d\}$ and $Y = \{1, 2, 3, 4\}$, respectively.

(b) Demonstrate the Particle Swarm Optimization (PSO) algorithm. [5]

4. (a) i) Compare and contrast among Mamdani FIS, Sugeno FIS, and Tsukamoto FIS. [5]

ii) Discover (Using Mamdani FIS) the crisp output derived by using the Centroid of area (COA) of the overall production MF generated by the MFs *trapezoid(x;5, 10, 40, 50)* and *triangle(x;20, 30, 80)*.

(b) Examine Genetic algorithm with the maximization problem of $f(x) = (31 - x)^2$, where integer $x \in [0, 31]$. Assume {5, 13, 17, 29} as the initial set of solutions. [5]

5. (a) (i) Demonstrate Roulette wheel selection operation used in the Genetic algorithm (GA) with an example. Discuss the limitation of the selection procedure with an example. [5]

(ii) Demonstrate two-point and multi-point crossover techniques in binary coded GA.

(b) Outline Artificial Neural Network (ANN). Illustrate a 3-3-2 ANN structure and explain the relationship between its input and output. [5]

6. (a) (i) Define T-norm operators Minimum (*min*), Algebraic product (*ap*), Bounded product (*bp*), and Drastic product (*dp*). [5]

(ii) Prove the following identity by considering two fuzzy sets $\bar{A} = \frac{0.3}{x_1} + \frac{0.8}{x_3} + \frac{1.0}{x_4}$ and $\bar{B} = \frac{0.1}{x_2} + \frac{0.7}{x_4}$, defined in the universe of discourse $\bar{X} = \{x_1, x_2, x_3, x_4\}$:

$T_{dp}(a, b) \leq T_{bp}(a, b) \leq T_{ap}(a, b) \leq T_{min}(a, b)$,
where $a \in \bar{A}$ and $b \in \bar{B}$.

(b) (i) Show the flowchart of Ant Colony Optimization (ACO) and briefly discuss its components. [5]

(ii) Illustrates an example to show the path selection in ACO.

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